Fundamentals of Set Theory

Sets, Set Builder Form, Types, Operations, Counting, and Hasse Diagrams

What Is a Set?

A Set is a well-defined collection of distinct objects, called Elements or Members.

$$A = \{1, 2, 3, 4\}$$

- $2 \in A$ means that 2 is an element of A.
- $7 \notin A$ means that 7 is not an element of A.
- Order does not matter: $\{1, 2, 3\} = \{3, 2, 1\}$.
- Repeated elements are counted only once: $\{1, 1, 2\} = \{1, 2\}$.

Ways to Represent a Set

1. **Roster (listing) form**

$$V = \{a, e, i, o, u\}$$

2. **Descriptive form**

V is the Set of vowels in the English alphabet.

3. **Set-builder form**

$$V = \{x \mid x \text{ is a vowel in the English alphabet}\}$$

The symbol $\mid$ means "such that."

Set-Builder Form

Set-builder notation describes a Set by stating the property its elements must satisfy.

$$A = \{x \in \mathbb{Z} \mid -2 \leq x \leq 2\}$$

This means: A contains all integers from -2 to 2.

$$A = \{-2, -1, 0, 1, 2\}$$

General pattern

$$\{x \in U \mid P(x)\}$$

where U is the universe of possible values and $P(x)$ is the condition that an element must satisfy.

More Set-Builder Examples

- Even natural numbers:

$$E = \{x \in \mathbb{N} \mid x = 2k \text{ for some } k \in \mathbb{N}\}.$$

- Positive integers less than 6:

$$B = \{x \in \mathbb{Z} \mid 0 < x < 6\} = \{1, 2, 3, 4, 5\}.$$

- Squares of the first five natural numbers (starting at 1):

$$S = \{n^2 \mid n \in \mathbb{N},\ 1 \leq n \leq 5\} = \{1, 4, 9, 16, 25\}.$$

Common Types of Sets

Empty set A set with no elements: $\varnothing$ or $\{\}$.

Singleton set A set with exactly one element: $\{0\}$.

Finite set A set with a limited number of elements: $\{a, b, c\}$.

Infinite set A set with infinitely many elements: $\mathbb{N} = \{1, 2, 3, \ldots\}$.

Equal sets Sets containing exactly the same elements.

Equivalent sets Sets having the same number of elements.

More Types: Subsets and Disjoint Sets

- A is a subset of B, written $A \subseteq B$, if every element of A is also in B.

$$\{1, 2\} \subseteq \{1, 2, 3\}.$$

- A proper subset, $A \subsetneq B$, is a subset that is not equal to B.
- Two sets are disjoint if they have no common elements:

$$A \cap B = \varnothing.$$

- The power set $\mathcal{P}(A)$ is the set of all subsets of A. If $A = \{a, b\}$, then

$$\mathcal{P}(A) = \{\varnothing, \{a\}, \{b\}, \{a, b\}\}.$$

Basic Set Operations

Let $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$.

Union Elements in A, in B, or in both:

$$A \cup B = \{1, 2, 3, 4, 5\}.$$

Intersection Elements common to both:

$$A \cap B = \{3\}.$$

Difference Elements in the first set but not the second:

$$A \setminus B = \{1, 2\}, \qquad B \setminus A = \{4, 5\}.$$

Complement and Key Laws

If the universal set is U, the complement of A is

$$A^c = U \setminus A.$$

Important identities:

$$A \cup \varnothing = A \qquad A \cap U = A$$
$$A \cup A = A \qquad A \cap A = A$$
$$A \cup A^c = U \qquad A \cap A^c = \varnothing$$

De Morgan's laws

$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$

Inclusion–Exclusion Principle

The inclusion–exclusion principle prevents elements in an overlap from being counted more than once.
For two finite sets:

$$\boxed{|A \cup B| = |A| + |B| - |A \cap B|}$$

Example
In a class, 18 students study Mathematics, 12 study Physics, and 5 study both. The number studying at least one subject is

$$|M \cup P| = 18 + 12 - 5 = \boxed{25}.$$

Inclusion–Exclusion for Three Sets

For three finite sets:

$$\begin{aligned}|A \cup B \cup C| = |A| + |B| + |C| \\ - |A \cap B| - |A \cap C| - |B \cap C| \\ + |A \cap B \cap C|.\end{aligned}$$

Why add the triple intersection back?

The elements in all three sets are:

- counted three times in the first line;
- subtracted three times in the second line;
- therefore counted zero times so far.

Adding the triple intersection once gives each such element a final count of 1.

Partially Ordered Sets

A partial order is a relation $\preceq$ that is:

- reflexive: $a \preceq a$;
- antisymmetric: if $a \preceq b$ and $b \preceq a$, then $a = b$;
- transitive: if $a \preceq b$ and $b \preceq c$, then $a \preceq c$.

A set together with a partial order is called a partially ordered set (or poset).
Example: the positive divisors of 6, ordered by divisibility:

$$D = \{1, 2, 3, 6\}, \qquad a \preceq b \iff a \mid b.$$

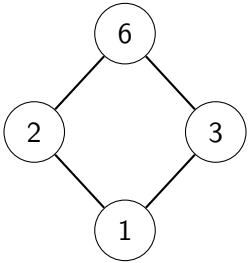
What Is a Hasse Diagram?

A Hasse diagram is a simplified drawing of a finite partially ordered set.

- Smaller elements are placed lower and larger elements higher.
- A line joins a to b when b covers a.
- Reflexive loops and edges implied by transitivity are omitted.
- The diagram represents the order relation, not necessarily numerical size.

Example Hasse Diagram: Divisors of 6

The order is divisibility. Thus $1 \mid 2$, $1 \mid 3$, $2 \mid 6$, and $3 \mid 6$.



Neither 2 nor 3 is above the other: they are incomparable under divisibility.

How to Construct a Hasse Diagram

1. List the elements of the poset.
2. Determine which pairs are related by the partial order.
3. Remove reflexive relations.
4. Remove relations that follow from transitivity.
5. Place lower elements below higher elements and draw the remaining cover relations.

Tip: For divisibility, connect a to b when $a \mid b$ and there is no listed element strictly between them in the divisibility order.

Quick Review

- A set is a well-defined collection of distinct elements.
- Set-builder notation describes elements through a condition.
- Common set types include empty, finite, infinite, subset, and power set.
- Union, intersection, difference, and complement are basic operations.
- Inclusion–exclusion counts unions without double-counting overlaps.
- Hasse diagrams display cover relations in a partially ordered set.

Practice Questions

1. Write $\{2, 4, 6, 8, 10\}$ in set-builder form.
2. If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5\}$, find $A \cup B$, $A \cap B$, and $A \setminus B$.
3. In a group, 20 people like tea, 15 like coffee, and 7 like both. How many like tea or coffee?
4. Draw the Hasse diagram for $\{1, 2, 4, 8\}$ ordered by divisibility.